H4 environmental-variable selection

Step 4: outcome-blind PC1 representative per category (max |PC1 loading|; Jolliffe 1972). Heat terms kept for the five equations. Domain-wise LASSO retained as secondary screening.

	UCM			Atmo raw	Atmo average	LYS1 raw	LYS2 raw	LYS average	GVI imagery	SVF imagery	Sound		Solar	Empatica		Eye-tracker	Questionnaire	Metadata / QC			
STEP 1 **Inventory (208)**	AH AIR RH AIR T bot AIR T mid AIR T top AIR temp fp GPS speed kmh HUMIDEX HUMIDEX SR IO flag IR back IR down IR front	IR left IR right IR spot gnd IR spot left IR spot right IR spot sky IR up MRT MRT S OPT PET PET SR SND dBA SUN alt	SUN az SUN Gh UTCI UTCI SR WIND AWA WIND AWS WIND TWD WIND TWS	left humidity left pm1 left pm10 left pm2.5 left temperature right humidity right pm1 right pm10 right pm2.5 right temperature	pm1 avg pm10 avg pm2.5 avg	LYS1 lux cat LYS1 b' LYS1 g' LYS1 kelvin LYS1 lux LYS1 medi LYS1 movement LYS1 r' LYS1 rgbb LYS1 rgbg LYS1 rgbir LYS1 rgbr	LYS2 lux cat LYS2 b' LYS2 g' LYS2 kelvin LYS2 lux LYS2 medi LYS2 movement LYS2 r' LYS2 rgbb LYS2 rgbg LYS2 rgbir LYS2 rgbr	kelvin avg lux avg medi avg	Grass phase mean GVI 50m GVI grass 50m GVI phase mean GVI plant 50m GVI tree 50m Plant phase mean Tree phase mean	Building % Combined Vegetation % Grass % Plant % Sky % SVF Holmer Tree % Wall %	band proportion 1000-4000 Hz band proportion 125-1000 Hz band proportion 20-125 Hz band proportion 4000-8000 Hz band proportion 8000-20000 Hz dominant frequency (Hz) octave proportion 1000 Hz octave proportion 125 Hz octave proportion 2000 Hz octave proportion 250 Hz octave proportion 31.5 Hz octave proportion 4000 Hz octave proportion 500 Hz octave proportion 63 Hz	octave proportion 8000 Hz spectral centroid (Hz) spectral rolloff 85% (Hz)	Clearness Index Clearness Index G0 Clearness Index Infer	acc x acc y acc z eda phasic eda scl usiemens eda tonic heart rate hrv fd hf hrv fd hfnu hrv fd lf hrv fd lf hf ratio hrv fd lfnu hrv fd total power hrv fd vlf	hrv td rmssd hrv td sdnn met skin temperature vector magnitude	blink duration s distance from center eyelid aperture avg fixation duration s fixation rate gaze centrality gaze dispersion gaze velocity pupil change rate pupil diameter avg saccade amplitude saccade duration s saccade peak velocity stress composite	current comfort 1 7 environmental comfort mean 1 5 pleasantness 1 5 prs11 being away mean 1 5 prs11 coherence mean 1 5 prs11 fascination mean 1 5 prs11 scope mean 1 5 prs11 total mean 1 5 space familiarity 1 5 stai6 mean 1 5 thermal comfort 1 5	age bike frequency bmi category who bmi kg m2 clothing accessories clothing footwear clothing height raw clothing lower body clothing skin tone clothing upper body clothing weight raw commute modes daily walking Date	Date (Atmo/LYS) Date (eye) Date (UCM) Datetime gender GPS alt GPS hdg GPS HDOP GPS lat GPS lon GPS speed valid sample count GVI match dist m has children height m	HRV 5min available HRV 5min max gap (s) HRV 5min removed pct HRV 5min segment end HRV 5min segment start HRV 5min valid rri count HRV 5min valid span (s) MAG hdg name nationality own bike for experiment participant id ParticipantID pct complete	phase start offset s PhaseID regular physical activity residence type residential area response index site familiarity 1 10 vision correction weight kg years in brussels
STEP 2 **Representatives (15)** **Rules + coverage**	AIR RH AIR temp HUMIDEX	MRT PET SND dBA	SUN Gh UTCI WIND AWS		pm1 avg pm10 avg pm2.5 avg			kelvin avg lux avg	GVI 50m	SVF											

	Acoustic	Thermal	Index	Radiation	Pollution	Light	Greenery	Unverified
STEP 3 **Domain Grouping (15)**	SND dBA	AIR temp AIR RH WIND AWS MRT	HUMIDEX PET UTCI	SUN Gh SVF	pm1 avg pm2.5 avg pm10 avg	lux avg kelvin avg	GVI 50m	
STEP 4 **PC1 representative** **per category** **(+ heat terms)**	**Acoustic** **Combined Empatica + Eyetracker** SND dBA Empatica n=1132; ocular n=1725 windows	**Thermal** **Combined Empatica + Eyetracker** AIR temp AIR RH WIND AWS Empatica n=1132; ocular n=1725 windows	**Index** **Combined Empatica + Eyetracker** HUMIDEX PET UTCI Empatica n=1132; ocular n=1725 windows	**Radiation** **Combined Empatica + Eyetracker** MRT SUN Gh Empatica n=1132; ocular n=1725 windows	**Pollution** **Combined Empatica + Eyetracker** pm2.5 avg Empatica n=1132; ocular n=1707 windows	**Light** **Combined Empatica + Eyetracker** lux avg Empatica n=1132; ocular n=1707 windows	**Greenery** **Combined Empatica + Eyetracker** GVI 50m Empatica n=1131; ocular n=1715 windows	**Unverified** **Combined Empatica + Eyetracker** Not entered

STEP 5
Within-person term grid
(5 equations; grouped term order)

Eq 1 · AIR_temp + AIR_RH + WIND_AWS — thermal × sound

sound + (humidity x sound) + (air temperature x sound) + (apparent wind x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | air temperature + humidity + apparent wind | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

Eq 2 · MRT — thermal × sound

sound + (mean radiant temperature x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | mean radiant temperature | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

Eq 3 · HUMIDEX — thermal × sound

sound + (HUMIDEX x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | HUMIDEX | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

Eq 4 · PET — thermal × sound

sound + (PET x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | PET | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

Eq 5 · UTCI — thermal × sound

sound + (UTCI x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | UTCI | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

STEP 6
GVIF/VIF diagnostic (threshold 5)
(no automatic predictor deletion)

5 exposure equations (one per thermal representation, each crossed with sound); generated from the catalog. Sum-to-zero PhaseID mains (WalkG, WalkU, BikeG deviations; BikeU implicit) in every equation. Within-bout acute exposures (standardized within-bout deviation); Step 6 GVIF/VIF diagnostic only (no automatic predictor deletion; heart-rate design max VIF-equivalent = 2.45).

Eq 1 · AIR_temp + AIR_RH + WIND_AWS — thermal × sound

sound + (humidity x sound) + (air temperature x sound) + (apparent wind x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | air temperature + humidity + apparent wind | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

GVIF diagnostic (HR n=2101): max VIF-equiv=2.21 (passes); PhaseID block VIF-equiv=1.06; no auto-removal

Eq 2 · MRT — thermal × sound

sound + (mean radiant temperature x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | mean radiant temperature | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

GVIF diagnostic (HR n=2101): max VIF-equiv=2.29 (passes); PhaseID block VIF-equiv=1.00; no auto-removal

Eq 3 · HUMIDEX — thermal × sound

sound + (HUMIDEX x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | HUMIDEX | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

GVIF diagnostic (HR n=2101): max VIF-equiv=1.16 (passes); PhaseID block VIF-equiv=1.01; no auto-removal

Eq 4 · PET — thermal × sound

sound + (PET x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | PET | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

GVIF diagnostic (HR n=2101): max VIF-equiv=2.25 (passes); PhaseID block VIF-equiv=1.00; no auto-removal

Eq 5 · UTCI — thermal × sound

sound + (UTCI x sound) | global solar radiation + GVI + lux + atmo_pm2.5_avg + exertion (vector magnitude) | UTCI | Phase: WalkG vs mean of phases + Phase: WalkU vs mean of phases + Phase: BikeG vs mean of phases

GVIF diagnostic (HR n=2101): max VIF-equiv=2.45 (passes); PhaseID block VIF-equiv=1.00; no auto-removal